IMPERIAL COLLEGE LONDON

EE4-29
EE9-CS3-2
EE9-FPN1-04

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING
EXAMINATIONS 2020

MSc and EEE/EIE PART IV: MEng and ACGI

OPTIMIZATION

Thursday, 30 April 10:00 am

Time allowed: 3:00 hours

There are FOUR questions on this paper.

Answer FOUR questions.

All questions carry equal marks

Any special instructions for invigilators and information for candidates are on page 1.

Examiners responsible First Marker(s) : A. Astolfi
Second Marker(s) : G. Scarciotti

© Imperial College London

OPTIMISATION

1. Consider the function

$$f(x) = x^3 - 2x + 2.$$

One wishes to use Newton's method to determine a solution of the equation $f(x) = 0$.

a) Plot the graph of the function $y = f(x)$ and show that the equation $f(x) = 0$ has a solution in the interval $x \in [-2, -1]$. Explain why this is the case. [2 marks]

b) Write Newton's iteration for the solution of the equation $f(x) = 0$. [2 marks]

c) Let $x_0 = 0$ and show that the Newton's iteration does not converge. In particular show that the sequence $\{x_k\}$ is such that $x_{2k} = x_{2k+2} = 0$ and $x_{2k+1} = x_{2k+3} = 1$, for all $k \geq 0$. [2 marks]

d) To explain the behaviour observed in part c) consider again the Newton's iteration derived in part b) and write x_{k+2} as a function of x_k, that is determine a relation of the form

$$x_{k+2} = F(x_k).$$

Show that $0 = F(0)$ and $1 = F(1)$, that is 0 and 1 are *fixed points* of the map F.

(Hint: do not simplify the expression of F.) [6 marks]

e) Consider now the quadratic equation

$$x^2 - c = 0,$$

with $c > 0$.

i) Write Newton's iteration for the solution of this equation. [2 marks]

ii) Exploiting the Newton's iteration in part e.i) write x_{k+2} as a function of x_k, that is determine a relation of the form

$$x_{k+2} = G(x_k).$$

Show that the unique fixed points of G, that is the points $\bar{x}$ such that $\bar{x} = G(\bar{x})$, are $\bar{x} = \pm\sqrt{c}$. Hence show that the unique fixed points of the iteration $x_{k+2} = G(x_k)$ are the solutions of the equation $x^2 - c = 0$. Finally argue that the *periodic behaviour* observed for the cubic equation cannot occur for quadratic equations. [6 marks]

2. Consider an optimization problem with objective function, to be minimized,

$$f(x,y) = x^2 y$$

and decision variables x and y such that the constraint

$$g(x,y) = x^2 + y^2 = 1$$

holds.

Hint: in what follows you may exploit the fact that the problem is symmetric with respect to x, that is the problem does not change if one replaces x with $-x$.

a) Plot the level lines of the objective function and the admissible set. Show that all points are regular points for the constraint. [2 marks]

b) Write the necessary conditions of optimality for the problem and determine candidate optimal solutions. [6 marks]

c) Using second order sufficient conditions of optimality classify the candidate optimal solutions determined in part b), that is determine which one is a local (strict) minimizer and which one is a local (strict) maximizer. [4 marks]

d) Compute the value of the objective function at the local minimizers and at the local maximizers determined in part c). Show that there is a local minimizer and a local maximizer for which the value of the objective function is the same. Hence, determine the global maximizers and minimizers. [4 marks]

e) Consider the change of variables $x = \cos\theta$ and $y = \sin\theta$. Show that the resulting transformed problem is an unconstrained optimization problem. Plot the objective function as a function of θ and verify the conclusions of parts c) and d). [4 marks]

3. Consider three devices that send data over a common link of capacity $C > 0$ bits/second. Let $r = (r_1, r_2, r_3)$ be the vector of data rates selected for the three devices.

Consider the *weighted proportionally fair rate allocation problem* described by the equations (note the maximization *sign*):

$$\max_{r_1,r_2,r_3} \ln r_1 + 2\ln r_2 + 3\ln r_3,$$
$$r_1 + r_2 + r_3 \leq C,$$
$$r_1 \geq 0,\ r_2 \geq 0,\ r_3 \geq 0.$$

The weights imply that higher indexed devices have higher priority in the rate maximization.

a) Write first order necessary conditions of optimality, disregarding the positivity constraints for r_1, r_2 and r_3. [2 marks]

b) Using the first order necessary conditions of optimality determined in part a) compute a candidate optimal solution. Show that the candidate optimal solution satisfies the positivity constrains and it is such that the constraint $r_1 + r_2 + r_3 \leq C$ is satisfied with the equality sign. [4 marks]

c) Consider the problem

$$\max_{r_1,r_2,r_3} \ln r_1 + 2\ln r_2 + 3\ln r_3,$$
$$r_1 + r_2 + r_3 = C.$$

Write the exact augmented Lagrangian function

$$S(r,\lambda) = f(r) + \lambda' g(r) + \frac{1}{\varepsilon}\|g(r)\|^2 + \eta \left\| \frac{\partial g(r)}{\partial r} \nabla_r L(r,\lambda) \right\|^2,$$

with $r = (r_1, r_2, r_3)$, for this constrained optimization problem and show that the optimal solution determined in part b) is a stationary point of $S(r,\lambda)$. [6 marks]

d) Consider now the case of n devices that send data over a common link of capacity $C > 0$ bits/second. Let $(r_1, r_2, \cdots, r_n)$ be the vector of data rates selected for the devices. Consider the *weighted proportionally fair rate allocation problem* described by the equations (note, again, the maximization *sign*):

$$\max_{r_1,r_2,\cdots,r_n} \alpha_1 \ln r_1 + \alpha_2 \ln r_2 + \cdots + \alpha_n \ln r_n,$$
$$r_1 + r_2 + \cdots + r_n \leq C,$$
$$r_1 \geq 0,\ r_2 \geq 0,\ \cdots,\ r_n \geq 0,$$

with $0 < \alpha_1 < \alpha_2 < \cdots < \alpha_n$. Exploiting the result of part b) determine a candidate optimal solution for this problem. [8 marks]

4. The Frank-Wolfe method is a method to solve nonlinear programming problems of the form

$$\min_x f(x), \qquad Ax \leq b,$$

where f is convex and differentiable and the admissible set is bounded. The method consists of the following steps.

(S0) Select a feasible point x^0 and set $k = 0$.

(S1) Compute an optimal solution ξ of the linear programming problem (recall that ∇f is a column vector)

$$\min_x x^T \nabla f(x^k), \qquad Ax \leq b.$$

(S2) If $(\xi - x^k)^T \nabla f(x^k) = 0$ then STOP, else set $x^{k+1} = x^k + t_k(\xi - x^k)$, with $t_k \in [0,1]$ and such that it performs an exact line search along the direction $\xi - x^k$, that is t_k is the global minimizer of the function $f(x^{k+1})$.

In the Frank-Wolfe method the quantity $(\xi - x^k)^T \nabla f(x^k)$ is either zero or negative and it expresses the *optimality gap* between the original problem and the modified, linear, problem. In (S2) the line search parameter could be set to $\frac{1}{k+1}$ or to any $t_k \in [0,1]$. One can show that for any x^0 the sequence $\{x^k\}$ is bounded and any of its converging subsequences converges to a global solution of the original problem.

Consider now the problem

$$\min_x (x_1 - 3)^2 + (x_2 - 1)^2, \qquad 0 \leq x_1 \leq 2, \qquad 0 \leq x_2 \leq 2.$$

a) By plotting the admissible set and the level lines of the objective function determine the optimal solution of the problem. [2 marks]

b) Let $x^0 = (0,0)$.

i) Write the linear problem in (S1) and determine its optimal solution ξ. [4 marks]

ii) Show that $(\xi - x^0)^T \nabla f(x^0) < 0$. As indicated in (S2), to determine t_0 one has to perform an exact line search along the direction $\xi - x^0$, with $t_0 \in [0,1]$. Perform such an exact line search and show that $t_0 = 1$. Hence, determine x^1. [4 marks]

c) Let $x^1 = (2,2)$. (Note that this should be the point determined in part b.ii).)

i) Write the linear problem in (S1) and determine its optimal solution ξ. [4 marks]

ii) Show that $(\xi - x^1)^T \nabla f(x^1) < 0$ and that the optimality gap has been reduced, that is it is smaller in absolute value to the optimality gap for the previous step. As indicated in (S2), to determine t_1 one has to perform an exact line search along the direction $\xi - x^1$, with $t_1 \in [0,1]$. Perform such an exact line search and show that $t_1 = 1/2$. Hence, determine x^2. [4 marks]

d) One could show that the next iteration of the Frank-Wolfe algorithm yields $\xi = (2,1)$. Use this fact to show that for this value of ξ the optimality gap $(\xi - x^2)^T \nabla f(x^2)$ is zero, hence x^2 is the optimal solution of the original problem. Compare this result with the result of part a). [2 marks]